Profile comparison

2 profiles from C:\APPS\FBENCH\

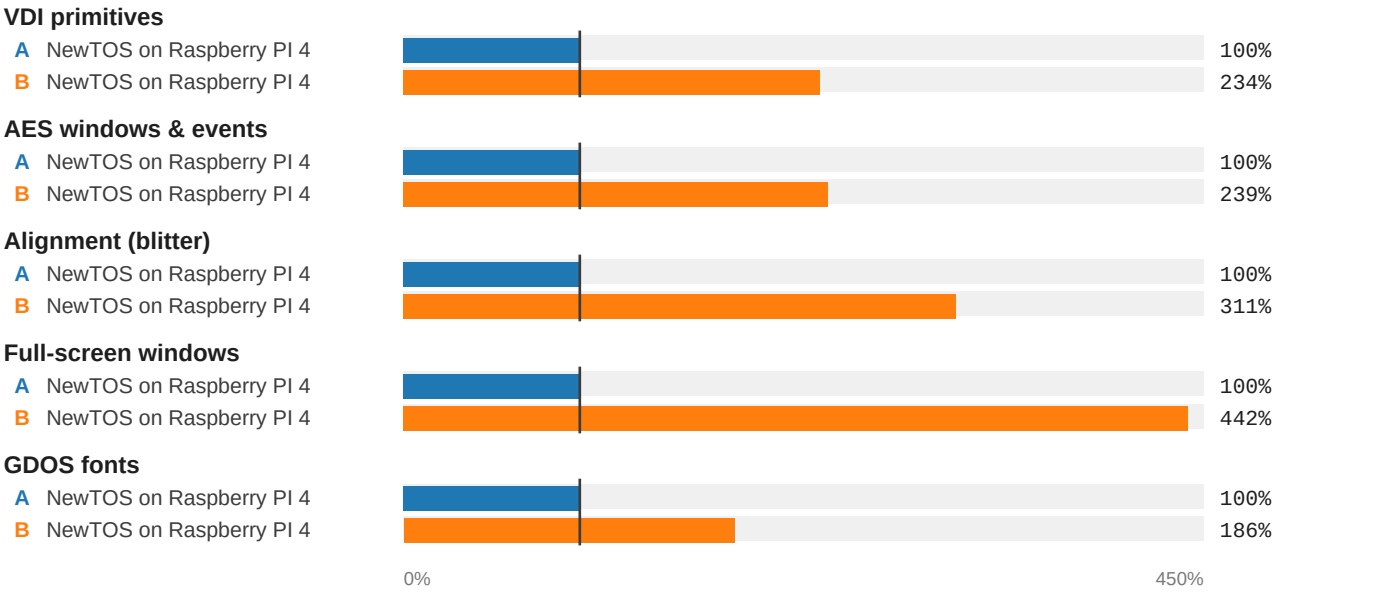
Machines

	Name / file	Screen	Planes	Test window	Desktop	Blitter	Date
A	**NewTOS on Raspberry PI 4**	1920x1080	16	1276x997	1920x1061	none	2026-11-06 00..
	MACHINE.FBP - unknown AArch64 TOS 4.04 1920x1080 256c bli						
B	**NewTOS on Raspberry PI 4**	1920x1080	16	1276x997	1920x1061	none	2026-11-08 00..
	OPTIMIZE.FBP - unknown AArch64 TOS 4.04 1920x1080 256c bl						

Machine A (NewTOS on Raspberry PI 4) is the reference: percentages are that machine's speed as 100%. Bold green marks the best machine for each test.

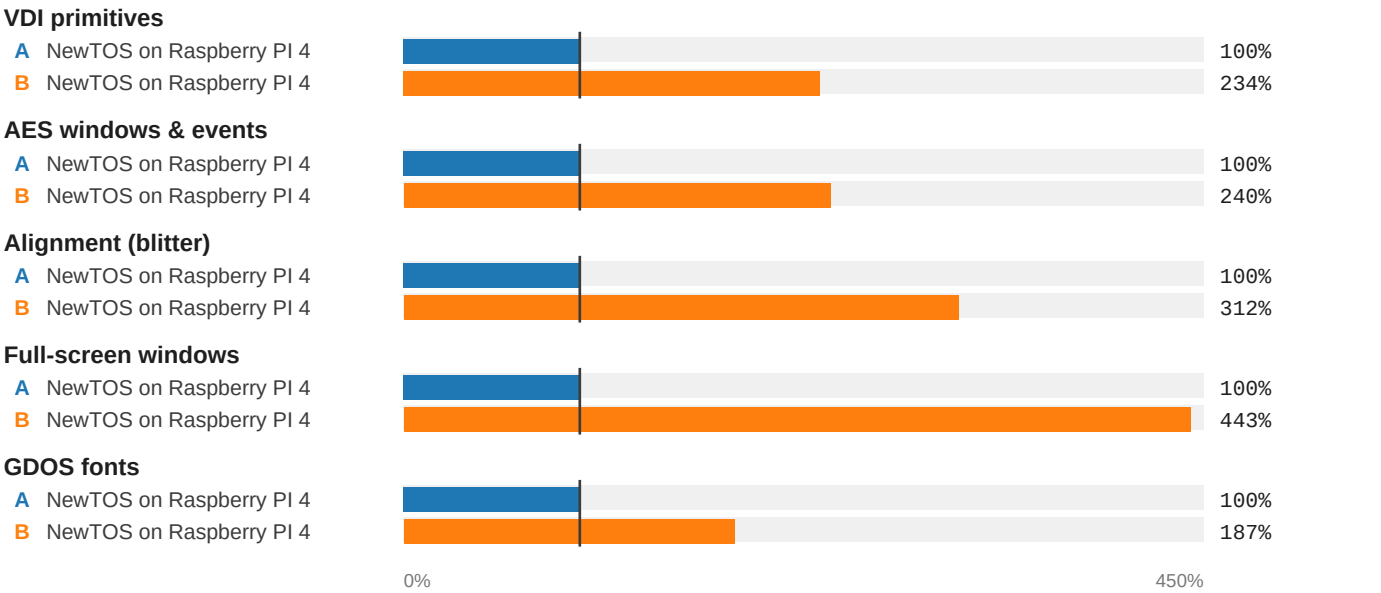
Summary - raw speed (% of A)

Average of the per-test percentages in each group, exactly as measured. A machine running a smaller window or fewer colours does less work per operation and looks faster here.



Summary - normalised (% of A)

The same averages after adjusting each test for the work one operation does on that machine: colour depth (bit planes), the size of the test window, and the desktop size for full-screen tests. This compares graphics throughput independent of the screen mode.



Raw results (operations per second)

VDI primitives

	A	B
Line, 2 points	22414	**23516**
Polyline, 32 points	763.1	**791.4**
Wide line (7px, round)	3327.0	**4034.4**
Dashed line	20572	**21152**
User line style	21765	**22649**
XOR line	11263	**20572**
Markers x16	22754	**23743**
Filled rect 32x32	37235	**91871**
Filled rect, full window	35.4	**578.2**
Bar, hatched + border	21977	**53843**
Pattern fill rect	37750	**90601**
User pattern fill	37925	**91444**
XOR filled rect	28108	**119300**
Polygon, 8 points	13985	**17553**
Filled circle	682.5	**2496.5**
Filled ellipse	892.0	**2870.0**
Arc	20858	**21251**
Pie slice	765.8	**2559.0**
Elliptical arc	21870	**22649**
Elliptical pie	1007.8	**2924.7**
Rounded box	39766	**44830**
Filled rounded box	3168.5	**9989.2**
Contour fill 64x40	4265.7	**7278.6**
Text, 15 chars	17225	**30912**
Text, 84 chars	2967.1	**5388.5**
Text with effects	7455.3	**19411**
Text rotated 90	15905	**31206**
Justified text	10477	**19139**
Text extent query	383624	**399984**
Blit scr->scr 64x64	22414	**38823**
Blit scroll window	119.2	**240.5**
Blit mem->scr 64x64	**27971**	**27971**
Blit scr->mem 64x64	**24765**	**24765**
Mono blit (vrt_cpyfm)	15282	**45055**
Get pixel	481881	**506557**
Hide/show mouse	20858	**111346**
Attribute calls x3	165342	**172462**
Open/close workstation	46151	**47034**

AES windows & events

	A	B
Window create+delete	135279	**146721**
Window open+close	26.2	**227.6**
Window move	335.8	**713.4**
Window resize	198.0	**909.0**
Window top (2 windows)	52.1	**460.5**
Set window title	860.5	**1576.3**
Set info line	864.7	**1583.8**
Set slider pos+size	1247.8	**2425.5**
Scroll window content	112.7	**218.6**
wind_get work area	**338447**	321253
Rectangle list walk	**148270**	145995
wind_calc	280867	**303406**
wind_update pair	**169223**	167649
objc_draw dialog	3621.6	**4150.4**

objc_find	**287718**	282211
form_dial redraw 64x48	4305.6	**11206**
Menu bar redraw	3975.7	**5935.2**
graf_mkstate	**340044**	322836
graf_mouse off+on	19786	**87063**
Event poll (evnt_multi)	137204	**141105**
Message round trip	**61535**	59361

Alignment (blitter)

	A	B
Fill 64x64, aligned	11376	**115923**
Fill 64x64, x+3	10934	**46368**
Blit 64x64, aligned	24093	**57343**
Blit 64x64, src+3 dst+11	21152	**35445**
Mem->scr 64x64, aligned	**28108**	**28108**
Mem->scr 64x64, x+3	**27835**	27701
Mono blit, aligned	15282	**47260**
Mono blit, x+3	15132	**44830**
Text, x on 8 px	17981	**35445**
Text, x+3	16383	**29824**
Scroll up, aligned edges	123.9	**221.8**
Scroll up, edges x+5	123.9	**209.3**
Scroll left 16 px	124.5	**186.3**
Scroll left 3 px	123.9	**185.4**
Move window, aligned	320.5	**635.8**
Move window, off-grid	317.4	**620.3**
Open/close win, aligned	25.9	**225.4**
Open/close win, off-grid	25.9	**223.0**
Scroll window, aligned	347.5	**776.6**
Scroll window, off-grid	342.4	**730.4**

Full-screen windows

	A	B
Open/close full window	13.1	**134.2**
Full <-> normal size	26.0	**241.7**
Top full / test window	26.7	**278.6**
Fill full window	23.3	**377.3**
Scroll up 1 line	77.4	**176.8**
Scroll left 16 px	76.6	**181.0**
Scroll left 8 px	79.0	**130.0**
Scroll left 3 px	79.8	**124.5**
Scroll text 8pt	80.1	**173.2**
Scroll text 10pt	77.4	**176.8**
Scroll text 12pt	77.4	**176.8**
Scroll text 14pt	77.4	**176.8**
Scroll text 18pt	76.6	**176.0**
Scroll text 24pt	72.4	**184.5**
Scroll text 36pt	72.4	**184.5**
Scroll text 48pt	72.4	**184.5**
Scroll, size per line	75.5	**178.5**

GDOS fonts

	A	B
Text 6pt	37066	**47952**
Text 8pt	37066	**48187**
Text 10pt	12986	**23293**
Text 12pt	12923	**23183**
Text 14pt	12986	**23516**
Text 18pt	6761.3	**16061**
Text 24pt	3478.6	**8652.5**

Text 36pt	3495.6	**8693.3**
Text 48pt	3512.7	**8734.5**
Text 72pt	3461.8	**8572.0**
Text 10pt bold	12737	**22754**
vst_point, 8 sizes	381762	**399984**
Scroll, mixed sizes	108.2	**220.7**
Text extent at 12pt	383624	**399984**

Relative to A (raw, %)

VDI primitives

	A	B
Line, 2 points	100%	**104%**
Polyline, 32 points	100%	**103%**
Wide line (7px, round)	100%	**121%**
Dashed line	100%	**102%**
User line style	100%	**104%**
XOR line	100%	**182%**
Markers x16	100%	**104%**
Filled rect 32x32	100%	**246%**
Filled rect, full window	100%	**1633%**
Bar, hatched + border	100%	**244%**
Pattern fill rect	100%	**240%**
User pattern fill	100%	**241%**
XOR filled rect	100%	**424%**
Polygon, 8 points	100%	**125%**
Filled circle	100%	**365%**
Filled ellipse	100%	**321%**
Arc	100%	**101%**
Pie slice	100%	**334%**
Elliptical arc	100%	**103%**
Elliptical pie	100%	**290%**
Rounded box	100%	**112%**
Filled rounded box	100%	**315%**
Contour fill 64x40	100%	**170%**
Text, 15 chars	100%	**179%**
Text, 84 chars	100%	**181%**
Text with effects	100%	**260%**
Text rotated 90	100%	**196%**
Justified text	100%	**182%**
Text extent query	100%	**104%**
Blit scr->scr 64x64	100%	**173%**
Blit scroll window	100%	**201%**
Blit mem->scr 64x64	**100%**	**100%**
Blit scr->mem 64x64	**100%**	**100%**
Mono blit (vrt_cpyfm)	100%	**294%**
Get pixel	100%	**105%**
Hide/show mouse	100%	**533%**
Attribute calls x3	100%	**104%**
Open/close workstation	100%	**101%**

AES windows & events

	A	B
Window create+delete	100%	**108%**
Window open+close	100%	**868%**
Window move	100%	**212%**
Window resize	100%	**459%**
Window top (2 windows)	100%	**883%**
Set window title	100%	**183%**

Set info line	100%	**183%**
Set slider pos+size	100%	**194%**
Scroll window content	100%	**193%**
wind_get work area	**100%**	94%
Rectangle list walk	**100%**	98%
wind_calc	100%	**108%**
wind_update pair	**100%**	99%
objc_draw dialog	100%	**114%**
objc_find	**100%**	98%
form_dial redraw 64x48	100%	**260%**
Menu bar redraw	100%	**149%**
graf_mkstate	**100%**	94%
graf_mouse off+on	100%	**440%**
Event poll (evnt_multi)	100%	**102%**
Message round trip	**100%**	96%

Alignment (blitter)

	A	B
Fill 64x64, aligned	100%	**1019%**
Fill 64x64, x+3	100%	**424%**
Blit 64x64, aligned	100%	**238%**
Blit 64x64, src+3 dst+11	100%	**167%**
Mem->scr 64x64, aligned	**100%**	**100%**
Mem->scr 64x64, x+3	**100%**	99%
Mono blit, aligned	100%	**309%**
Mono blit, x+3	100%	**296%**
Text, x on 8 px	100%	**197%**
Text, x+3	100%	**182%**
Scroll up, aligned edges	100%	**179%**
Scroll up, edges x+5	100%	**168%**
Scroll left 16 px	100%	**149%**
Scroll left 3 px	100%	**149%**
Move window, aligned	100%	**198%**
Move window, off-grid	100%	**195%**
Open/close win, aligned	100%	**870%**
Open/close win, off-grid	100%	**861%**
Scroll window, aligned	100%	**223%**
Scroll window, off-grid	100%	**213%**

Full-screen windows

	A	B
Open/close full window	100%	**1024%**
Full <-> normal size	100%	**929%**
Top full / test window	100%	**1043%**
Fill full window	100%	**1619%**
Scroll up 1 line	100%	**228%**
Scroll left 16 px	100%	**236%**
Scroll left 8 px	100%	**164%**
Scroll left 3 px	100%	**156%**
Scroll text 8pt	100%	**216%**
Scroll text 10pt	100%	**228%**
Scroll text 12pt	100%	**228%**
Scroll text 14pt	100%	**228%**
Scroll text 18pt	100%	**229%**
Scroll text 24pt	100%	**254%**
Scroll text 36pt	100%	**254%**
Scroll text 48pt	100%	**254%**
Scroll, size per line	100%	**236%**

GDOS fonts

	A	B
Text 6pt	100%	129%
Text 8pt	100%	130%
Text 10pt	100%	179%
Text 12pt	100%	179%
Text 14pt	100%	181%
Text 18pt	100%	237%
Text 24pt	100%	248%
Text 36pt	100%	248%
Text 48pt	100%	248%
Text 72pt	100%	247%
Text 10pt bold	100%	178%
vst_point, 8 sizes	100%	104%
Scroll, mixed sizes	100%	203%
Text extent at 12pt	100%	104%

Normalised results (% of A)

Each test's operations per second are multiplied by the work one operation does on that machine, then shown as a percentage of machine A. The 'scales' column says what the work is proportional to:

planes - fixed-size drawing (64x64 blits, text, 32x32 fills): bit planes only. len - lines across the window: (width + height) / 2 x planes. min - arcs and rounded boxes sized from the window: min(width, height) x planes. dim² - filled shapes whose radius follows the window: min(w, h)² x planes. area - whole test-window work (fills, scrolls, window moves): width x height x planes. desk - full-screen windows: desktop width x height x planes. calls - AES/VDI calls with no pixel work (events, messages, queries): not adjusted.

VDI primitives

	scales	A	B
Line, 2 points	len	100%	105%
Polyline, 32 points	len	100%	104%
Wide line (7px, round)	len	100%	121%
Dashed line	len	100%	103%
User line style	len	100%	104%
XOR line	len	100%	183%
Markers x16	planes	100%	104%
Filled rect 32x32	planes	100%	247%
Filled rect, full window	area	100%	1633%
Bar, hatched + border	planes	100%	245%
Pattern fill rect	planes	100%	240%
User pattern fill	planes	100%	241%
XOR filled rect	planes	100%	424%
Polygon, 8 points	planes	100%	126%
Filled circle	dim²	100%	366%
Filled ellipse	dim²	100%	322%
Arc	min	100%	102%
Pie slice	dim²	100%	334%
Elliptical arc	min	100%	104%
Elliptical pie	dim²	100%	290%
Rounded box	min	100%	113%
Filled rounded box	dim²	100%	315%
Contour fill 64x40	planes	100%	171%
Text, 15 chars	planes	100%	179%
Text, 84 chars	planes	100%	182%
Text with effects	planes	100%	260%
Text rotated 90	planes	100%	196%
Justified text	planes	100%	183%
Text extent query	calls	100%	104%
Blit scr->scr 64x64	planes	100%	173%
Blit scroll window	area	100%	202%

Blit mem->scr 64x64	planes	**100%**	**100%**
Blit scr->mem 64x64	planes	**100%**	**100%**
Mono blit (vrt_cpyfm)	planes	100%	**295%**
Get pixel	calls	100%	**105%**
Hide/show mouse	calls	100%	**534%**
Attribute calls x3	calls	100%	**104%**
Open/close workstation	calls	100%	**102%**

AES windows & events

	scales	A	B
Window create+delete	calls	100%	**108%**
Window open+close	area	100%	**869%**
Window move	area	100%	**212%**
Window resize	area	100%	**459%**
Window top (2 windows)	area	100%	**884%**
Set window title	calls	100%	**183%**
Set info line	calls	100%	**183%**
Set slider pos+size	calls	100%	**194%**
Scroll window content	area	100%	**194%**
wind_get work area	calls	**100%**	95%
Rectangle list walk	calls	**100%**	98%
wind_calc	calls	100%	**108%**
wind_update pair	calls	**100%**	99%
objc_draw dialog	planes	100%	**115%**
objc_find	calls	**100%**	98%
form_dial redraw 64x48	planes	100%	**260%**
Menu bar redraw	calls	100%	**149%**
graf_mkstate	calls	**100%**	95%
graf_mouse off+on	calls	100%	**440%**
Event poll (evnt_multi)	calls	100%	**103%**
Message round trip	calls	**100%**	96%

Alignment (blitter)

	scales	A	B
Fill 64x64, aligned	planes	100%	**1019%**
Fill 64x64, x+3	planes	100%	**424%**
Blit 64x64, aligned	planes	100%	**238%**
Blit 64x64, src+3 dst+11	planes	100%	**168%**
Mem->scr 64x64, aligned	planes	**100%**	**100%**
Mem->scr 64x64, x+3	planes	**100%**	**100%**
Mono blit, aligned	planes	100%	**309%**
Mono blit, x+3	planes	100%	**296%**
Text, x on 8 px	planes	100%	**197%**
Text, x+3	planes	100%	**182%**
Scroll up, aligned edges	area	100%	**179%**
Scroll up, edges x+5	area	100%	**169%**
Scroll left 16 px	area	100%	**150%**
Scroll left 3 px	area	100%	**150%**
Move window, aligned	area	100%	**198%**
Move window, off-grid	area	100%	**195%**
Open/close win, aligned	area	100%	**870%**
Open/close win, off-grid	area	100%	**861%**
Scroll window, aligned	area	100%	**223%**
Scroll window, off-grid	area	100%	**213%**

Full-screen windows

	scales	A	B
Open/close full window	desk	100%	1024%
Full <-> normal size	desk	100%	930%
Top full / test window	desk	100%	1043%
Fill full window	desk	100%	1619%
Scroll up 1 line	desk	100%	228%
Scroll left 16 px	desk	100%	236%
Scroll left 8 px	desk	100%	165%
Scroll left 3 px	desk	100%	156%
Scroll text 8pt	desk	100%	216%
Scroll text 10pt	desk	100%	228%
Scroll text 12pt	desk	100%	228%
Scroll text 14pt	desk	100%	228%
Scroll text 18pt	desk	100%	230%
Scroll text 24pt	desk	100%	255%
Scroll text 36pt	desk	100%	255%
Scroll text 48pt	desk	100%	255%
Scroll, size per line	desk	100%	236%

GDOS fonts

	scales	A	B
Text 6pt	planes	100%	129%
Text 8pt	planes	100%	130%
Text 10pt	planes	100%	179%
Text 12pt	planes	100%	179%
Text 14pt	planes	100%	181%
Text 18pt	planes	100%	238%
Text 24pt	planes	100%	249%
Text 36pt	planes	100%	249%
Text 48pt	planes	100%	249%
Text 72pt	planes	100%	248%
Text 10pt bold	planes	100%	179%
vst_point, 8 sizes	calls	100%	105%
Scroll, mixed sizes	area	100%	204%
Text extent at 12pt	calls	100%	104%

Notes

Normalisation is a model, not a measurement: it assumes drawing cost grows with the number of bit planes touched and the pixels covered. Real hardware adds fixed per-call overheads, so small operations on high-resolution modes are usually under-rewarded and large ones fit best. GDOS text at the same point size covers different pixel counts at different screen resolutions (DPI); that is not adjusted. Profiles measured with different test durations, blitter settings or background programs are not directly comparable.